

Mathematical Methods in Uncertainty Quantification (MMUQ) – L01

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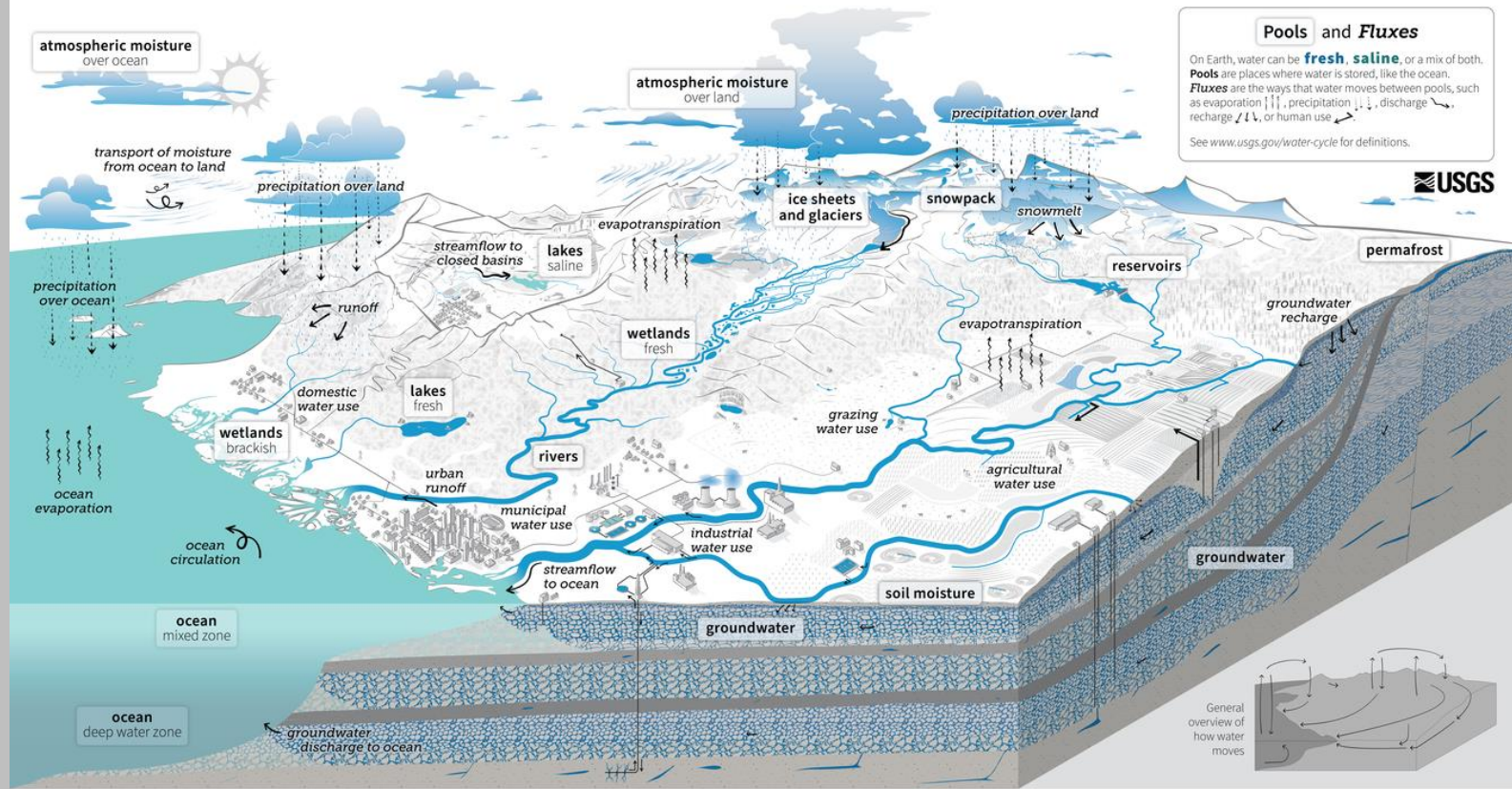
School of Engineering and Design

Chair of Hydrology and River Basin Management (Prof. Disse)

Organization: Moodle and Webdisk

MMUQ L01 – General Concepts

- The Water Cycle
- Rainfall-runoff Modeling
- Spatio-temporal Aspects
- Conceptual Models
- Efficiency Metrics
- Model Parameter Optimization
- Local and Global Optimizations



The Water Cycle

The water cycle describes where water is found on Earth and how it moves. Water can be stored in the atmosphere, on Earth's surface, or below the ground. It can be in a liquid, solid, or gaseous state. Water moves between the places it is stored at large scales and at very small scales. Water moves naturally and because of human interaction, both of which affect where water is stored, how it moves, and how clean it is.

Liquid water can be fresh, saline (salty), or a mix (brackish). Ninety-six percent of all water is saline and stored in **oceans**. Places like the ocean, where water is stored, are called **pools**. On land, saline water is stored in **saline lakes**, whereas fresh water is stored in liquid form in **freshwater lakes**, artificial **reservoirs**, **rivers**, **wetlands**, and in soil as **soil moisture**. Deeper underground, liquid water is stored as **groundwater** in aquifers, within the cracks and pores of rock. The solid, frozen form of water is stored in **ice sheets**, **glaciers**, and **snowpack** at high elevations or near the Earth's poles. Frozen water is also found in the soil as **permafrost**. Water vapor, the gaseous form of water, is stored as **atmospheric moisture** over the ocean and land.

As it moves, water can transform into a liquid, a solid, or a gas. The different ways in which water moves between pools are known as **fluxes**. **Circulation** mixes water in the oceans and transports water vapor in the atmosphere. Water moves between the atmosphere and the Earth's surface through **evaporation**, **evapotranspiration**, and **precipitation**. Water moves across the land surface through **snowmelt**, **runoff**, and **streamflow**. Through infiltration and **groundwater recharge**, water moves into the ground. When underground, groundwater flows within aquifers and can return to the surface through **springs** or from natural **groundwater discharge** into rivers and oceans.

Humans alter the water cycle. We redirect rivers, build dams to store water, and drain water from wetlands for development. We use water from rivers, lakes, reservoirs, and groundwater aquifers. We use that water (1) to supply our **homes and communities**; (2) for **agricultural** irrigation and **grazing** livestock; and (3) in **industrial** activities like thermoelectric power generation, mining, and aquaculture. The amount of available water depends on how much water is in each pool (water quantity). Water availability also depends on when and how fast water moves (water timing), how much water is used (water use), and how clean the water is (water quality).

Human activities affect **water quality**. In agricultural and urban areas, irrigation and precipitation wash fertilizers and pesticides into rivers and groundwater. Power plants and factories return heated and contaminated water to rivers. Runoff carries chemicals, sediment, and sewage into rivers and lakes. Downstream from these types of sources, contaminated water can cause harmful algal blooms, spread diseases, and harm habitats. **Climate change** is also affecting the water cycle. It affects water quality, quantity, timing, and use. Climate change is also causing ocean acidification, sea level rise, and extreme weather. Understanding these impacts can allow progress toward sustainable water use.

Rainfall-Runoff Modeling (RRM)

- Modeling subset of the Water Cycle
- What happens to rain?
 - Before Sea
 - Catchment scale
- Interactions on the way
- Results of interactions
- Need to predict
 - Design
 - Prepare
 - Warn

Spatio-temporal Aspects

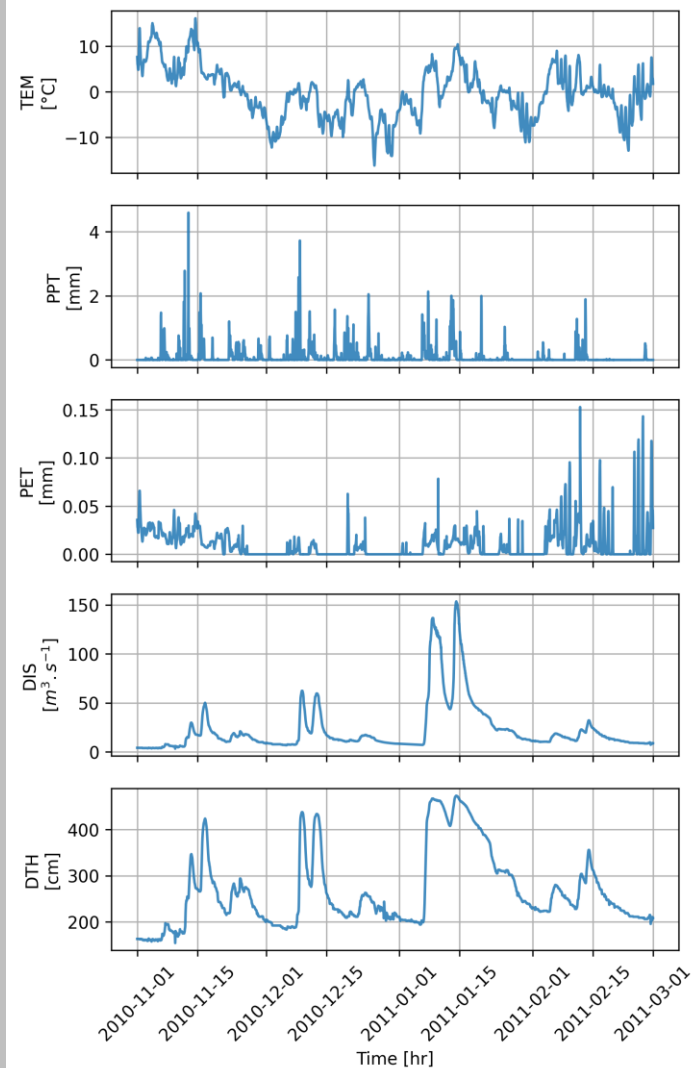
- RRM can be lumped or distributed
 - in space!
 - Definitions vary
- Distributed
 - Discretized catchment (cells)
 - Semi- or fully-distributed
 - Same parameters everywhere (semi)
 - Different parameters everywhere (fully)
- Lumped
 - Mean behavior in space
 - Single parameter vector
- Temporal resolutions can vary
 - e.g., daily or hourly
- Discretization in space-time affects output
- We will have
 - Lumped
 - Hourly

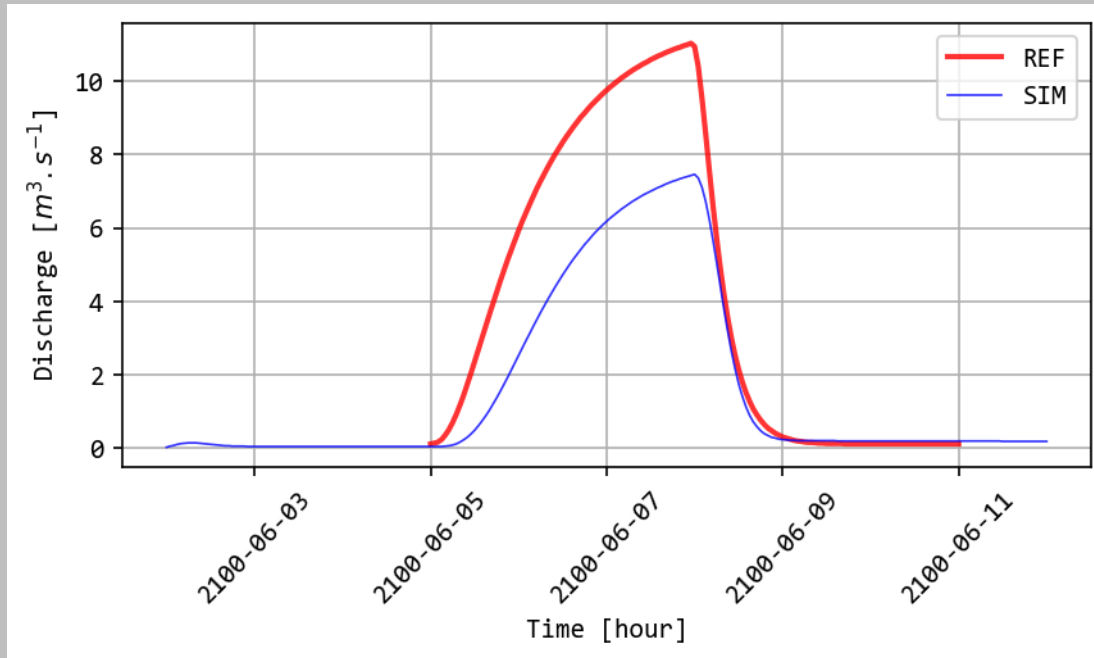
Conceptual Modeling

- One type of modeling
- Others?
 - Empirical, Physically-based, Machine Learning
- Conceptual == Statistical approximation
 - What happens on average (under given conditions)
- Used when catchments too large or not enough data
- Examples?
 - Snow melt per °C rise in air temperature instead of snow energy balance equations
 - Evapotranspiration dependent on soil moisture instead of actual evapotranspiration

Basic terms in RRM

- Temperature (TEM)
- Precipitation (PPT)
- Potential evapotranspiration (PET)
- River discharge (DIS)
- River flow depth (DTH)





Efficiency Metrics

- Based on what features you want to keep/reproduce
- Generally, related to the sum of the squared differences
- Can be anything you want
 - Mass balances (given time scale(s))
 - Specific features of the hydrograph
- Standard ones
 - RMSE
 - Nash-Sutcliffe
 - Log Nash-Sutcliffe
 - Kling-Gupta
- Mix-n-Match
- Drawbacks

Sum of Squared Errors (SSE)

$$SSE = \sum_i (OBS_i - SIM_i)^2$$

$$OBJ_{SSE} = SSE$$

- Squared differences
 - Goal is to have the least possible value
 - Always positive
 - Best is zero
 - Worst is infinity
- Same form as an objective function
 - Needs to be made as small as possible

Nash-Sutcliffe Efficiency (NSE)

$$NSE = 1 - \frac{\sum_i (OBS_i - SIM_i)^2}{\sum_i (OBS_i - \overline{OBS})^2}$$

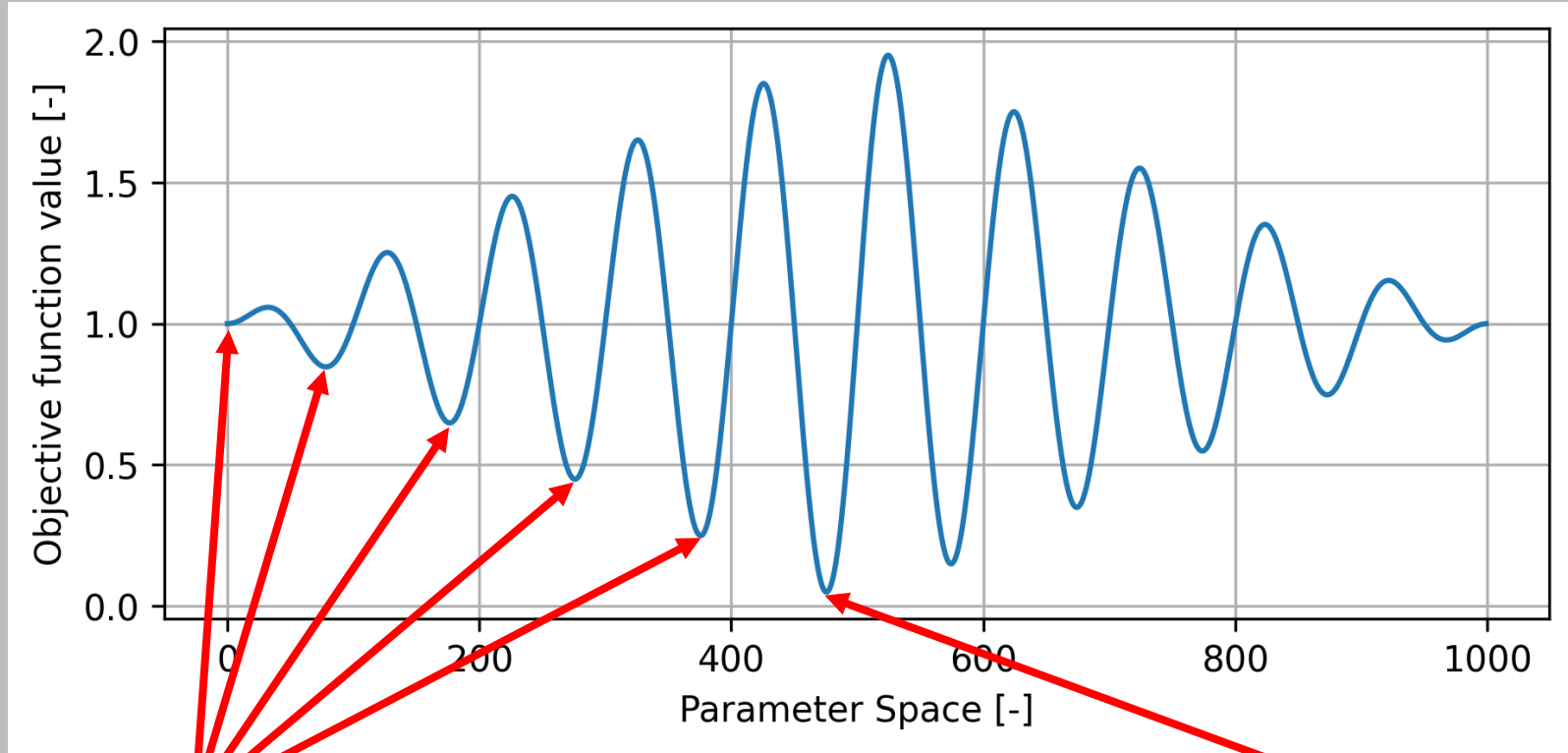
- Normalized squared differences
 - Mean flow used as a prediction
- Best is 1.0
- Worst is negative infinity
- Zero means error same as if using mean

$$OBJ_{NSE} = 1 - NSE$$

- Different form as an objective function
- Needs to be made as small as possible

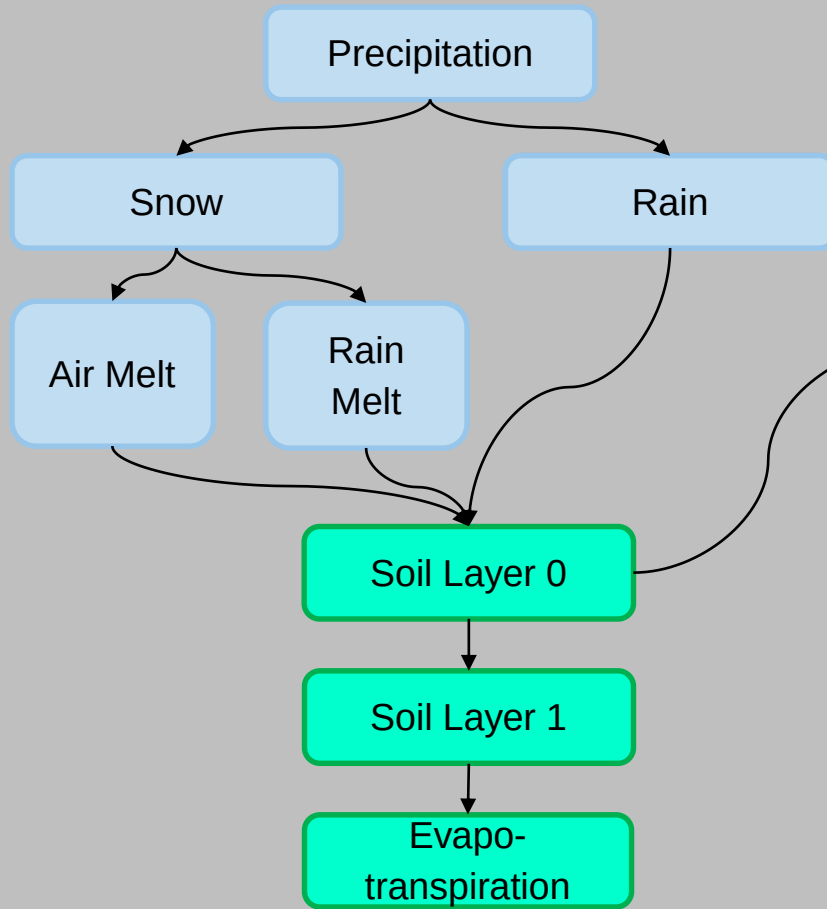
Model Parameter Optimization

- Parameters control behavior
 - Accuracy
- Parameter measurement
 - Not possible for conceptual
- Adjustment by using known data
 - Inputs
 - Outputs
- Fit via efficiency metrics
- Objective function
 - Modified efficiency metric(s)
 - Ideally, a zero
 - The lower, the better
- Local and Global Optima

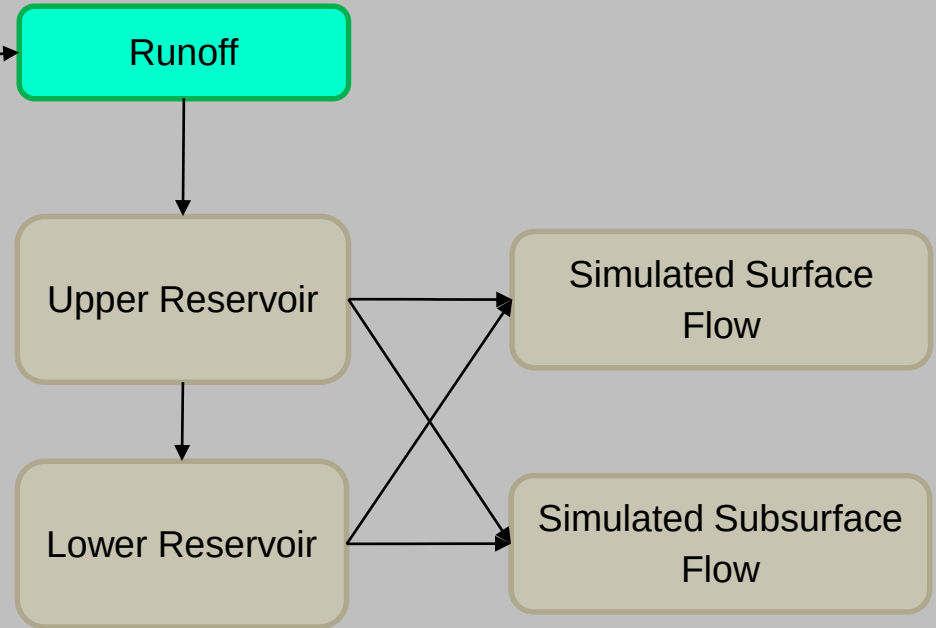


Local

Global



Lumped HBV (hbv1d012a)



That was it for today, thanks!